

JEE MAIN REVISION MASTER: COMPOUNDS CONTAINING NITROGEN

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 229 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Compounds Containing Nitrogen - Core Application	229 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Compounds Containing Nitrogen - Core Application (Very High Yield — 229 PYQs)

Core Principle: Key problem pattern in Compounds Containing Nitrogen.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Which statements are True? A. In Hoffmann bromamide degradation, 4 moles of NaOH and 2 moles of Br₂ are consumed per mole of an amide B..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following compounds according to increasing order of boiling points. $n\text{-C}_4\text{H}_9\text{OH}$ (A), $n\text{-C}_4\text{H}_9\text{NH}_2$ (B), $n\text{-C}_4\text{H}_{10}$ (C)..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Product C of the following reaction sequence will be..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Arrange the following compounds according to increasing order of boiling points.

$\text{n-C}_4\text{H}_9\text{OH}$ (A), $\text{n-C}_4\text{H}_9\text{NH}_2$ (B), $\text{n-C}_{10}\text{H}_{22}$ (C) and $\text{C}_5\text{H}_{12}\text{N}_2$ (D).

- (A) $\text{C} < \text{B} < \text{A} < \text{D}$
- (B) $\text{D} < \text{C} < \text{B} < \text{A}$
- (C) $\text{C} < \text{D} < \text{B} < \text{A}$
- (D) $\text{D} < \text{B} < \text{A} < \text{C}$

Correct Answer: None

Solution Analysis: (A) $\text{n-C}_4\text{H}_9\text{OH}$ (an alcohol) forms strong intermolecular hydrogen bonding between O-H groups of different molecules. Because of this strong attraction, more heat is needed to separate the molecules, so its boiling point is the highest. (B) $\text{n-C}_4\text{H}_9\text{NH}_2$ (a primary amine) also shows intermolecular hydrogen bonding due to the presence of N-H bonds. But nitrogen is less electronegative than oxygen, so N-H hydrogen bonding is weaker than O-H hydrogen bonding. Hence, its boiling point is lower than the alcohol. (C) $\text{n-C}_{10}\text{H}_{22}$ is a non-polar hydrocarbon. It cannot form hydro